A Proof that $f(r,s,p)$ is Increasing on $[0,1]$

Note: this document can definitely be shortened considerably; I just wrote everything out to make sure that I didn't make any simple mistakes.

We previously defined the polynomial $f(r,s,p)$ using the formula

$$f(r,s,p) = \frac{a(r,s,p)}{a(r,s,p) + b(r,s,p)}.$$

And we already know that

$$\begin{aligned} a(r,s,p) &= p^s \left[a(r-1,s,p) + b(r-1,s,p)\right] \\ b(r,s,p) &= (1-p)^r \left[a(r,s-1,p) + b(r,s-1,p)\right] \end{aligned}$$

Proposition 1.

$$f(r,s,p) = 1 - \frac{1 - \left(\frac{p}{1-p}\right)^s}{1 - \left(\frac{p}{1-p}\right)^{r+s}}, \tag{1}$$

or, equivalently,

$$f(r,s,p) = \frac{p^s (1-p)^r - p^{r+s}}{(1-p)^{r+s} - p^{r+s}}.$$

To prove this, we will first need to know a few things about Gaussian q-binomial coefficients.

Definition 2. The Gaussian q-binomial coefficient is defined by

$$\binom{n}{k}_q = \frac{(1-q^n)(1-q^{n-1})\cdots(1-q^{n-k+1})}{(1-q)(1-q^2)\cdots(1-q^k)}.$$

Moreover, we know that the following recursive property holds for the Gaussian q-binomial coefficients:

Proposition 3.

$$\binom{n}{k}_q = q^k \binom{n-1}{k}_q + \binom{n-1}{k-1}_q.$$

Lemma 4.

$$\begin{aligned} a(r,s,p) &= \binom{r+s-1}{s}_{\frac{p}{1-p}} p^s (1-p)^{(r-1)s} \\ b(r,s,p) &= \binom{r+s-1}{s-1}_{\frac{p}{1-p}} (1-p)^{rs} \end{aligned}$$

Proof. We proceed by induction. First note that $\binom{1+1-1}{1}_{\frac{p}{1-p}} p^1 (1-p)^{(1-1)1} = p = a(1,1,p)$ and $\binom{1+1-1}{1-1}_{\frac{p}{1-p}} (1-p)^{1\cdot 1} = 1-p = b(1,1,p)$. Now, for a fixed s, assume that the formulae for $a(r-1,s,p)$ and $b(r-1,s,p)$ hold. Then we may compute

$$
\begin{aligned}
a(r,s,p) &= p^s\left[a(r-1,s,p)+b(r-1,s,p)\right] \\
&= p^s\left[\binom{(r-1)+s-1}{s}_{\frac{p}{1-p}} p^s(1-p)^{(r-1-1)s} + \binom{(r-1)+s-1}{s-1}_{\frac{p}{1-p}}(1-p)^{(r-1)s}\right] \\
&= p^s\left[\left(\frac{p}{1-p}\right)^s\binom{(r+s-1)-1}{s}_{\frac{p}{1-p}}(1-p)^{(r-1)s} + \binom{(r+s-1)-1}{s-1}_{\frac{p}{1-p}}(1-p)^{(r-1)s}\right] \\
&= \left[\left(\frac{p}{1-p}\right)^s\binom{(r+s-1)-1}{s}_{\frac{p}{1-p}} + \binom{(r+s-1)-1}{s-1}_{\frac{p}{1-p}}\right]p^s(1-p)^{(r-1)s} \\
&= \binom{r+s-1}{s}_{\frac{p}{1-p}} p^s(1-p)^{(r-1)s}.
\end{aligned}
$$

Likewise, for a fixed r, if we assume that the formulae for $a(r,s-1,p)$ and $b(r,s-1,p)$ hold, then we may compute

$$
\begin{aligned}
b(r,s,p) &= (1-p)^r\left[a(r,s-1,p)+b(r,s-1,p)\right] \\
&= (1-p)^r\left[\binom{r+(s-1)-1}{s-1}_{\frac{p}{1-p}} p^{s-1}(1-p)^{(r-1)(s-1)} + \binom{r+(s-1)-1}{s-1}_{\frac{p}{1-p}}(1-p)^{r(s-1)}\right] \\
&= (1-p)^r\left[\left(\frac{p}{1-p}\right)^{s-1}\binom{(r+s-1)-1}{s-1}_{\frac{p}{1-p}}(1-p)^{r(s-1)} + \binom{(r+s-1)-1}{r}_{\frac{p}{1-p}}(1-p)^{r(s-1)}\right] \\
&= \left[\left(\frac{p}{1-p}\right)^{s-1}\binom{(r+s-1)-1}{s-1}_{\frac{p}{1-p}} + \binom{(r+s-1)-1}{s-1}_{\frac{p}{1-p}}\right](1-p)^r(1-p)^{r(s-1)} \\
&= \binom{r+s-1}{s-1}_{\frac{p}{1-p}}(1-p)^{rs}.
\end{aligned}
$$

♣

Proof of Proposition 1. First we will observe that

$$
\begin{aligned}
a(r,s,p)+b(r,s,p) &= \binom{r+s-1}{s}_{\frac{p}{1-p}} p^s(1-p)^{(r-1)s} + \binom{r+s-1}{s-1}_{\frac{p}{1-p}}(1-p)^{rs} \\
&= \left(\frac{p}{1-p}\right)^s\binom{r+s-1}{s}_{\frac{p}{1-p}}(1-p)^{rs} + \binom{r+s-1}{s-1}_{\frac{p}{1-p}}(1-p)^{rs} \\
&= \left[\left(\frac{p}{1-p}\right)^s\binom{r+s-1}{s}_{\frac{p}{1-p}} + \binom{r+s-1}{s-1}_{\frac{p}{1-p}}\right](1-p)^{rs} \\
&= \left[\binom{r+s}{s}_{\frac{p}{1-p}}(1-p)^{rs}\right] \\
&= \frac{a(r+1,s,p)}{p^s}.
\end{aligned}
$$

So we may write

$$
\begin{aligned}
f(r,s,p) &= \frac{a(r,s,p)}{a(r,s,p)+b(r,s,p)} \\
&= \frac{a(r,s,p)\,p^s}{a(r+1,s,p)} \\
&= \frac{\binom{r+s-1}{s}_{\frac{p}{1-p}} p^s\,(1-p)^{(r-1)s}\,p^s}{\binom{r+s}{s}_{\frac{p}{1-p}} p^s\,(1-p)^{rs}} \\
&= \frac{\binom{r+s-1}{s}_{\frac{p}{1-p}} (1-p)^{(r-1)s}\,p^s}{\binom{r+s}{s}_{\frac{p}{1-p}} (1-p)^{rs}} \\
&= \frac{\binom{r+s-1}{s}_{\frac{p}{1-p}} \left(\frac{p}{1-p}\right)^s}{\binom{r+s}{s}_{\frac{p}{1-p}}} \\
&= \frac{\frac{\left(1-\left(\frac{p}{1-p}\right)^{r+s-1}\right)\left(1-\left(\frac{p}{1-p}\right)^{r+s-2}\right)\cdots\left(1-\left(\frac{p}{1-p}\right)^{r}\right)}{\left(1-\frac{p}{1-p}\right)\left(1-\left(\frac{p}{1-p}\right)^{2}\right)\cdots\left(1-\left(\frac{p}{1-p}\right)^{s}\right)}}{\frac{\left(1-\left(\frac{p}{1-p}\right)^{r+s}\right)\left(1-\left(\frac{p}{1-p}\right)^{r+s-1}\right)\cdots\left(1-\left(\frac{p}{1-p}\right)^{r+1}\right)}{\left(1-\frac{p}{1-p}\right)\left(1-\left(\frac{p}{1-p}\right)^{2}\right)\cdots\left(1-\left(\frac{p}{1-p}\right)^{s}\right)}} \left(\frac{p}{1-p}\right)^s \\
&= \frac{1-\left(\frac{p}{1-p}\right)^r}{1-\left(\frac{p}{1-p}\right)^{r+s}} \left(\frac{p}{1-p}\right)^s \\
&= 1-\frac{1-\left(\frac{p}{1-p}\right)^s}{1-\left(\frac{p}{1-p}\right)^{r+s}}.
\end{aligned}
$$

♣

Proposition 5. $f(r,s,p)$ is strictly increasing on $[0,1]$.

Proof. The argument here is that the derivative of $f(r,s,p)$ never vanishes on $[0,1]$. Taking the derivative of (1) with respect to p, we get

$$
\frac{\partial}{\partial p}\left[f(r,s,p)\right] = \frac{\left(\frac{p}{1-p}\right)^{s-1}\left(\frac{1}{1-p}\right)^2\left(s\left(1-\left(\frac{p}{1-p}\right)^{r+s}\right)-(r+s)\left(\frac{p}{1-p}\right)^r\left(1-\left(\frac{p}{1-p}\right)^s\right)\right)}{\left(1-\left(\frac{p}{1-p}\right)^{r+s}\right)^2}.
$$

Which simplifies to

$$
\frac{\partial}{\partial p}\left[f(r,s,p)\right] = \frac{\left(\frac{p}{1-p}\right)^{s-1}\left(\frac{1}{1-p}\right)^2\left(s-(r+s)\left(\frac{p}{1-p}\right)^r+r\left(\frac{p}{1-p}\right)^{r+s}\right)}{\left(1-\left(\frac{p}{1-p}\right)^{r+s}\right)^2}.
$$

For the sake of simplicity, let $g(r,s,p,t)$ represent $\frac{\partial}{\partial p}\left[f(r,s,p)\right]$ with every instance of $(1-p)$ replaced by t. The following factorization then becomes apparent

$$\begin{aligned}
g(r,s,p,t) &= \frac{\left(\frac{p}{t}\right)^{s-1}\left(\frac{1}{t}\right)^{2}\left(s-(r+s)\left(\frac{p}{t}\right)^{r}+r\left(\frac{p}{t}\right)^{r+s}\right)}{\left(1-\left(\frac{p}{t}\right)^{r+s}\right)^{2}} \\
&= \frac{p^{s-1}\,t^{r-1}\,\left(s\,t^{r+s}-(r+s)\,p^{r}t^{s}+r\,p^{r+s}\right)}{\left(t^{r+s}-p^{r+s}\right)^{2}} \\
&= \frac{p^{s-1}\,t^{r-1}\,(p-t)^{2}\nu(r,s,p,t)}{(p-t)^{2}\,\delta(r,s,p,t)} \\
&= \frac{p^{s-1}\,t^{r-1}\,\nu(r,s,p,t)}{\delta(r,s,p,t)}
\end{aligned}$$

where

$$\begin{aligned}
\nu(r,s,p,t) &= r\sum_{i=1}^{s} i\,p^{r+s-1-i}\,t^{i-1}+s\sum_{i=1}^{r-1} i\,p^{i-1}\,t^{r+s-1-i} \\
\delta(r,s,p,t) &= \left(\sum_{i=0}^{r+s-1} p^{r+s-1-i}\,t^{i}\right)^{2}.
\end{aligned}$$

so, for example,

$$\nu(3,4,p,t) = 3p^5+6p^4t+9p^3t^2+12p^2t^3+8p\,t^4+4t^5,$$

and

$$\delta(3,4,p,t) = \left(p^6+p^5t+p^4t^2+p^3t^3+p^2t^4+p\,t^5+t^6\right)^2.$$

We can check the formula for $\nu(r,s,p,t)$ by simple multiplication:

$$\begin{aligned}
(p-t)^2\,\nu(r,s,p,t) &= (p-t)^2\left(r\sum_{i=1}^{s} i\,p^{r+s-1-i}\,t^{i-1}+s\sum_{i=1}^{r-1} i\,p^{i-1}\,t^{r+s-1-i-1}\right) \\
&= (p-t)\left(r\sum_{i=1}^{s} p^{r+s-i}\,t^{i+1}-s\sum_{i=1}^{r-1} p^{i-1}\,t^{r+s-i}\right) \\
&= s-(r+s)\,p^{r}t^{s}+r\,p^{r+s}.
\end{aligned}$$

It is then not too difficult to see that for $(p,t)\in[0,1]^2\setminus\{(0,0)\}$, we have that $\nu(r,s,p,t)$ and $\delta(r,s,p,t)$ are strictly positive since all of the coefficients are positive. Thus, $g(r,s,p,1-p)=\frac{\partial}{\partial p}\left[f\right]$ is never zero on $(0,1)$.

It is easily verifiable that $f(r,s,0)=0$ and $f(r,s,1)=1$, so $f(r,s,p)$ is strictly increasing on $[0,1]$. ♣

If

$$\widetilde{f}(r,s,\alpha) = 1 - \frac{1-\alpha^s}{1-\alpha^{r+s}}$$

then

$$\frac{\partial}{\partial p}\left[\widetilde{f}(r,s,\alpha(p))\right] = \frac{\left(s\,\alpha^{s-1} - (r+s)\,\alpha^{r+s-1} + r\,\alpha^{r+2s-1}\right)\alpha'}{\left(1-\alpha^{r+s}\right)^2}.$$

and since $\alpha' > 0$, we just need to show that $s\,\alpha^{s-1} - (r+s)\,\alpha^{r+s-1} + r\,\alpha^{r+2s-1} > 0$ for all positive α. Factoring the numerator, we obtain

$$s\,\alpha^{s-1} - (r+s)\,\alpha^{r+s-1} + r\,\alpha^{r+2s-1} = (1-\alpha)^2\,\mu(r,s,\alpha)$$

where

$$\mu(r,s,\alpha) = r\sum_{i=1}^{s} i\,\alpha^{r+2s-2-i} + s\sum_{i=1}^{r-1}(r-i)\,\alpha^{r+s-2-i}$$

and $\mu(r,s,\alpha) > 0$ for all positive α since all of the coefficients are positive.